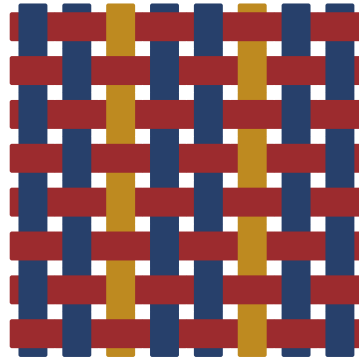




The Adversary's Loom

Notes on online classification & the Littlestone dimension

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

1 THE REPEATED GAME

We play in rounds $t = 1, \dots, T$. A **learner** must guess a hidden labelling; an **adversary** reveals the truth one bite at a time and is free to choose the worst continuation. Nothing is random. The only question is: how many times can the learner be *wrong*?

The picture to hold: the adversary is not a fixed distribution, it is a player watching you. Every bound below is a statement about a *tree of futures*, not about an average. If you remember one thing, remember the tree.

Definition 1.1 (*mistake bound*). A learner A for a class $\mathcal{H} \subseteq \{0, 1\}^{\mathcal{X}}$ has **mistake bound** M if, against every realisable sequence $(x_1, y_1), \dots, (x_T, y_T)$ with $y_t = h^*(x_t)$ for some $h^* \in \mathcal{H}$, it errs at most M times.

↔ Ldim

Definition 1.2 (*Littlestone dimension*). $\text{Ldim}(\mathcal{H})$ is the depth of the deepest *shattered* binary tree: a complete tree whose internal nodes carry points of $\mathcal{X}$ and whose every root-to-leaf branch is realised by some $h \in \mathcal{H}$.

The two definitions are secretly the same statement read from two sides. That is the whole content of the next knot.

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Is the deepest tree unique?  
Almost never.  
Does the SOA need to find it?  
No --- it only needs its *depth*.  
Revisit.

## 2 THE KNOT THAT TIES IT TOGETHER

**Theorem 2.1** (*Littlestone, 1988*). For every class  $\mathcal{H}$ , the optimal deterministic mistake bound equals  $\text{Ldim}(\mathcal{H})$ . The bound is achieved by the **Standard Optimal Algorithm** (SOA), which predicts the label keeping the larger residual Littlestone dimension.

*Proof. Upper bound.* Each time the SOA errs, the adversary's reply contradicts the majority side, so the surviving subclass drops its Littlestone dimension by at least one (this uses  $[\text{Ldim} \leftrightarrow]$ ). Starting from  $\text{Ldim}(\mathcal{H})$ , there can be at most that many mistakes.

*Lower bound.* Walk the shattered tree: at each node force the SOA to commit, then send the branch it did *not* choose. Every step is a mistake, and the tree is realisable by construction, so  $\text{Ldim}(\mathcal{H})$  mistakes are unavoidable. ■

■■■■■ I can rebuild the upper bound cold; the lower bound's "send the other branch" still feels like a rabbit from a hat.

**Example 2.2** (*thresholds on a line*). Let  $\mathcal{X} = [0, 1]$  and  $\mathcal{H}$  the thresholds  $h_a(x) = \mathbf{1}[x \geq a]$ . A shattered tree of depth  $k$  exists for every  $k$  (binary search keeps halving the live interval), so  $\text{Ldim} = \infty$ . *Online*, thresholds are unlearnable — yet their VC dimension is 1 and they are trivially PAC-learnable.

中文直觉：在线与离线的鸿沟，全藏在那棵「未来之树」里。VC 维只数一层标注的自由度，而 Littlestone 维数数的是对手能逼你走多深。阈值函数恰好是这道裂缝最干净的见证。

*Remark 2.3.* Randomisation softens but does not close the gap: the optimal expected mistake bound sits between  $\frac{1}{2} L_{\dim}$  and  $L_{\dim}$ , and the constant is itself a small open story.

The separation above is the seed of everything else in these notes: agnostic regret, bandit feedback, and the apple-tasting variants all measure how badly the tree can branch when the learner is handicapped.

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Open thread to pull next session: does a *margin* assumption collapse $L_{\dim}$ back down to something finite? Cf. the large-margin perceptron bound $1/\gamma^2$.